

Lecture 3

Last week: $P, P(x), \wedge, \exists, \forall$, Rules of Inference } logic.

This week: Proofs

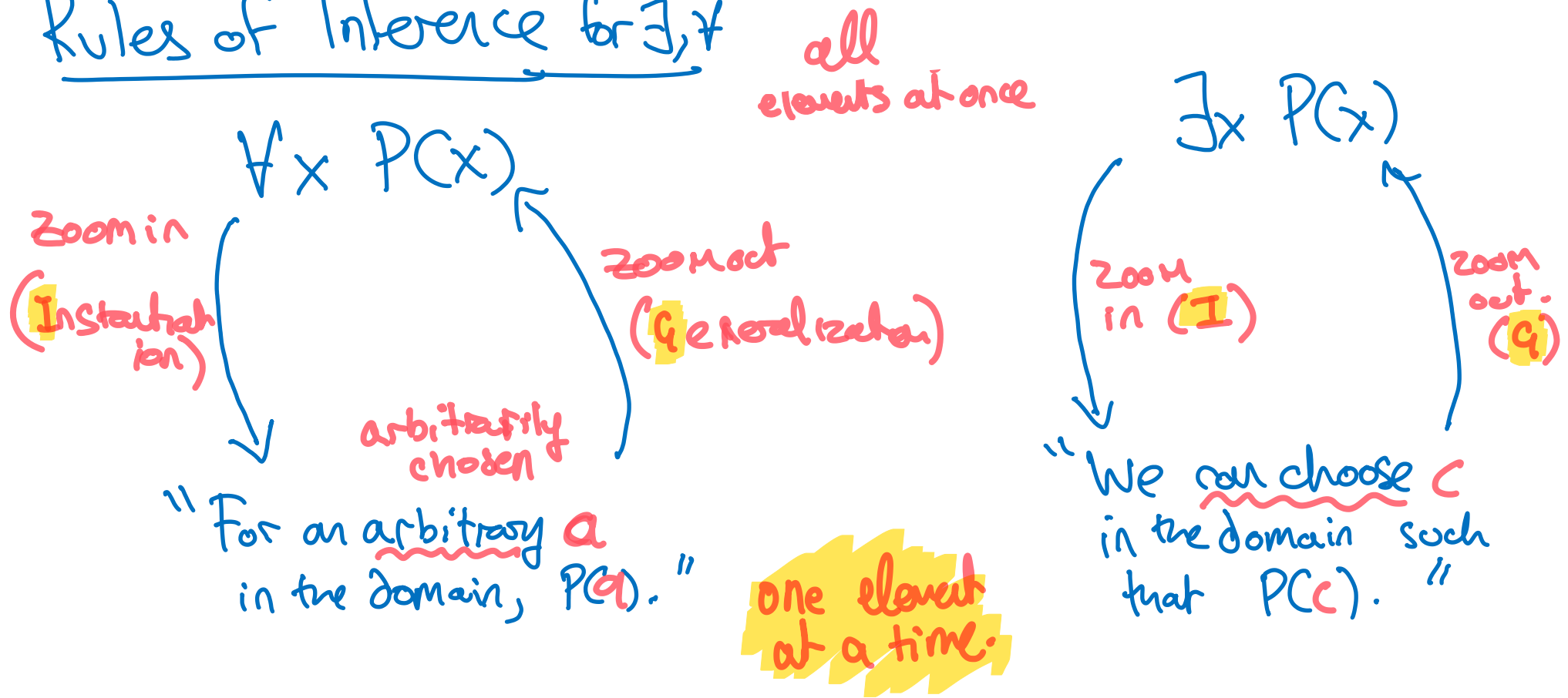
"Necessary Conclusions."
"Structured Writing."
"Absolute truth."

I Direct Proof

II Proof by Contrapositive

III Proof by contradiction

Rules of Inference for $\exists, \forall$

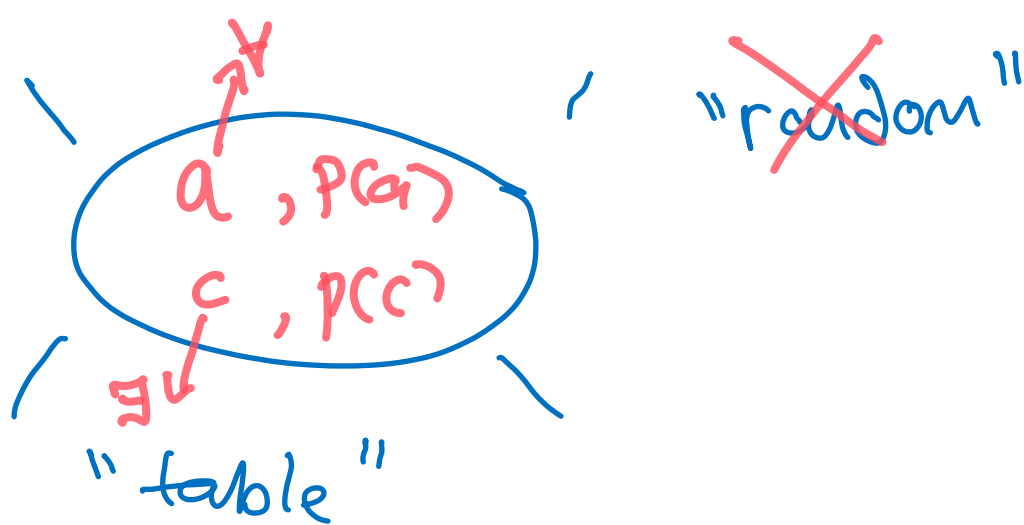


Metaphor #1: Zooming in and zooming out.

Metaphor #2 (I) Choosing an element and giving a name.

$\forall$: arbitrarily chosen
(by an adversary)

$\exists$: specifically chosen
(by you)



Defn: An argument is a sequence of propositions, each of which is a premise (i.e. an assumption) or a conclusion. It is valid if each conclusion follows from the previous propositions via rules of inference.

A proof is a valid argument used to establish the truth of a mathematical statement.

I Direct Proof. $[\forall x (P(x) \rightarrow Q(x))]$.

English

Defn: An integer n is even if there exists an integer k such that $n = 2k$.

Defn: An integer n is odd if there exists an integer k s.t. $n = 2k + 1$.

Prop A: If an integer n is odd, then n^2 is odd.

Logic

$\forall n$ (even(n) $\leftrightarrow$ $\exists k$ ($n = 2k$))
integers integers
(Defns are always $\leftrightarrow$)

$\forall n$ (odd(n) $\leftrightarrow$ $\exists k$ ($n = 2k + 1$))
defn

$\forall n$ (odd(n) $\rightarrow$ odd(n^2))

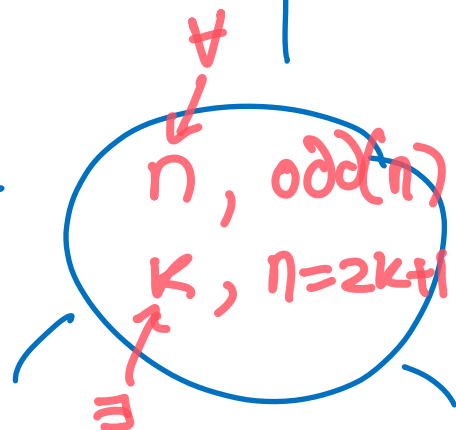
Prop 0: $\forall n$ (odd(n) $\leftrightarrow$ even(n))
<no proof>

Proof: Assume n is an arbitrary integer.

Assume n is odd.

By defn, we can choose k

s.t. $n = 2k + 1$.



Observe that

$$n^2 = (2k + 1)^2$$

$$= 4k^2 + 4k + 1$$

$$= 2(2k^2 + 2k) + 1.$$

integer

Thus, $n^2 = 2h + 1$ for $h = 2k^2 + 2k$,
so by definition n^2 is odd.

Since n is arbitrary, this concludes the proof.

Let n be arbitrary.

Assume n is odd. premise

$$\text{odd}(n) \leftrightarrow \exists k (n = 2k + 1) \text{ defn.}$$

Choose k s.t. $n = 2k + 1$. $(\exists I)$

//

$$n^2 = 2h + 1 \text{ for } h = 2k^2 + 2k, \text{ integer.}$$

$$\exists h (n^2 = 2h + 1) \quad (\exists I)$$

$$\therefore \text{odd}(n^2) \text{ defn.} \quad \therefore \forall n (\text{odd}(n) \rightarrow \text{odd}(n^2)). \quad (\forall I)$$

Poll: Find the mistakes in the following argument.

Theorem: For all integers n , if n^2 is odd then

Argument:

- ① Assume n is an ^{arbitrary} integer.
- ② Assume n^2 is odd.
- ③ Choose k such that $n^2 = 2k+1$.
- ④ Observe $n = \sqrt{2k+1}$. (what if? $k < 0$)
- ⑤ Notice
- ⑥ Thus,

premise

premise

n is odd.

Can be fixed by noticing $2k+1 = n^2 \geq 0$

defn odd. ($\exists k$)

Algebra.

So n is not even.

$\neg \text{even}(n) \rightarrow \text{odd}(n)$

$\forall \rightarrow n$

$\exists \rightarrow k$

$$n^2 = 2k+1$$

$$n \neq 2k$$

Fallacy

$$\neg \text{even}(n) \equiv \neg (\exists k (n=2k))$$

$$\equiv \forall k (n \neq 2k)$$

not justified.
"logical leap"

II Proof by contrapositive $\left[\forall x (P(x) \rightarrow Q(x)) \equiv \forall x (\neg Q(x) \rightarrow \neg P(x)) \right]$

Prop B: For every integer n , if n^2 is odd then n is odd.

We will show the contrapositive.

Assume n is an arbitrary integer.

Assume n is even.

By defn, we can choose k s.t. $n = 2k$.

Observe that $n^2 = 4k^2 = 2(2k^2)$.

Conclude n^2 is even. $\square$

$$\forall n (\text{odd}(n^2) \rightarrow \text{odd}(n)).$$

||| contrapositive.

$$\forall n (\neg \text{odd}(n) \rightarrow \neg \text{odd}(n^2)).$$

||| prop O

$$\forall n (\text{even}(n) \rightarrow \text{even}(n^2)).$$

⋮

$$\begin{array}{l} \forall - n, \text{even} \\ \exists - k, n = 2k \\ \quad n^2 = 4k^2 \end{array}$$

puts different things on the table

(cf. $n = \sqrt{2k+1}$ useless).

$$\therefore \forall n (\text{even}(n) \rightarrow \text{even}(n^2)). \quad (\forall n)$$

III Proof by Contradiction

Defn: A real number x is rational if there exist integers $p, q, q \neq 0$ s.t. $x = p/q$.

Otherwise it is irrational.

Fact: Can assume p, q have no common factors.
<will prove later>.

Corollary C: $\forall n (\text{odd}(n) \leftrightarrow \text{odd}(n^2))$.
By Propo: $\forall n (\text{even}(n) \leftrightarrow \text{even}(n^2))$.

$\forall x$ ^{real} (rational(x) $\leftrightarrow$ $\exists p, q (x = p/q, q \neq 0)$).

$\forall x$ (irrational(x) $\leftrightarrow$ $\forall p, q (q \neq 0 \rightarrow x \neq p/q)$).

Theorem: $\sqrt{2}$ is irrational.

Attempts:

"Assume $\sqrt{2}$ is irrational"

→ fallacy

"Assume p, q arbitrary
 $q \neq 0$ "

→ doesn't work

Proof: Assume $\sqrt{2}$ is rational. $(\neg P)$

rational($\sqrt{2}$). premise.

$\exists p, q (\sqrt{2} = p/q)$. defn.

Choose integers $p, q \neq 0$
s.t. $\sqrt{2} = p/q$.

$\sqrt{2}, p, q$ integers
 $\sqrt{2} = p/q$

Observe $2 = p^2/q^2$

which implies $p^2 = 2q^2$.

Thus, p^2 is even.

By Corollary C, p must be even.

By defn, we can choose k s.t. $p = 2k$.

Thus, $(2k)^2 = 4k^2 = 2q^2$.

Notice that $q^2 = 2k^2$.

Thus q^2 is even.

Thus q is even, by Corollary C.

This contradicts the Fact — p and q cannot both be even.

$\therefore \sqrt{2}$ is irrational $\square$

defn
invoke Corollary

$\exists$
 $\sqrt{2}, p, q, k$
 $2 = p^2/q^2$
 $p = 2k$
 q even

defn
Invoke
 F

To prove P
Assume $\neg P$
 $\vdots$
Derive F

 $\therefore P$

Creativity

logical structure $\square$

Summary

① Direct Proof

Goal

$$\forall x (P(x) \rightarrow Q(x))$$

Structure

Let x be arbitrary.
Assume $P(x)$

$$\vdots$$
$$Q(x)$$

$$\hline \therefore \forall x (P(x) \rightarrow Q(x))$$

② Contrapositive

$$\forall x (P(x) \rightarrow Q(x))$$

Let x be arbitrary.
Assume $\neg Q(x)$

$$\vdots$$

$$\neg P(x)$$

$$\forall x (\neg Q(x) \rightarrow \neg P(x))$$

$$\hline \forall x (P(x) \rightarrow Q(x)).$$

③ Contradiction

$$P \rightarrow Q$$

(or just Q)

Assume P
Assume $\neg Q$

$$\vdots$$

$$Q$$

$$\hline F$$

$$\therefore P \rightarrow Q$$
$$\hline (P \wedge \neg Q) \rightarrow F.$$